3 FBLA STUDENT ACTIVITIES

IST/CSP

Micro:bit Smart Hall Pass System

Instructions:

Assignment Overview

Your team has been hired by Centennial High School to design a **Smart Hall Pass System** using Micro:bits.

The goal is to replace a traditional paper hall pass with a digital wireless communication system.

Your final system must allow a student to:

- Request a hall pass
- Select the type of pass needed
- Send the request wirelessly
- Receive approval or denial from the teacher

Your final system must allow a teacher to:

- Receive requests
- View Student ID
- View Pass Type
- Approved requests
- Deny requests
- Track requests using enhancements

Team Structure

Student Device Team (2 Students)

Responsible for:

- Student ID
- Pass Type Selection
- Request Transmission
- Approval/Denial Displays

Teacher Device Team (2 Students)

Responsible for:

- Request Reception
- Approval System
- Statistics
- Dashboard Features

Timeline

Tuesday

Project Introduction and Starter Code

Wednesday

Student IDs and Pass Types

Thursday

Two-Way Communication

Friday

Enhancements and Final Testing

Monday

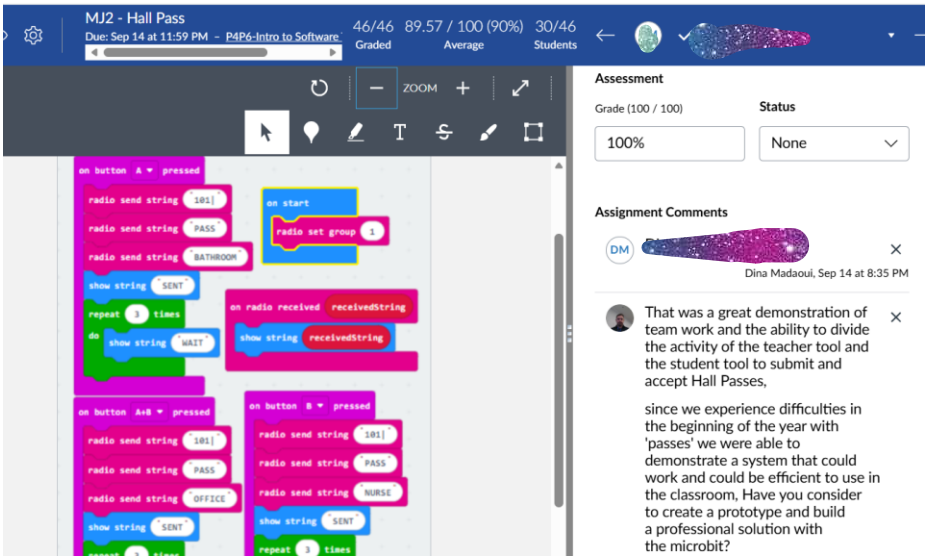
Whole-Class Live Demonstration

FBLA Event: Sales Presentation

Students design and build a Micro:bit-powered hall pass system intended to streamline student movement tracking and improve school operations. Students create a prototype, prepare a product pitch, and present the solution as a technology product.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback



AP Computer Science Principles

What's the Odds? Powerball vs. Bitcoin Mining

Instructions:

Scenario

A software company wants to create an application that compares two possible ways of making money with an initial investment of $1,000:

- Buying Powerball tickets
- Mining Bitcoin

Before programmers write code, they need researchers to determine:

- What information is required
- What assumptions must be made
- What mathematical concepts are needed
- What variables should be included
- What errors could occur

Your role is to act as a member of the software company's research team.

You are not coding during this phase of the project.

Driving Question

If you had $1,000 and a two-year time period, would you have a better chance of making money from:

- Powerball tickets
- Bitcoin mining

Support your conclusion using research and evidence.

Learning Objectives

Big Idea 1.1 Collaboration

- Work with classmates to discuss findings and ideas.
- Compare sources and information.
- Ask questions and provide feedback during the gallery walk.

Big Idea 1.2 Program Function and Purpose

- Determine what problem the future software should solve.
- Identify the purpose of the simulation.
- Determine what information users would want from the program.

Big Idea 1.3 Program Design and Development

- Identify variables needed before coding begins.

- Research required data sources.
- Determine what calculations and inputs the software would need.

Big Idea 1.4 Identifying and Correcting Errors

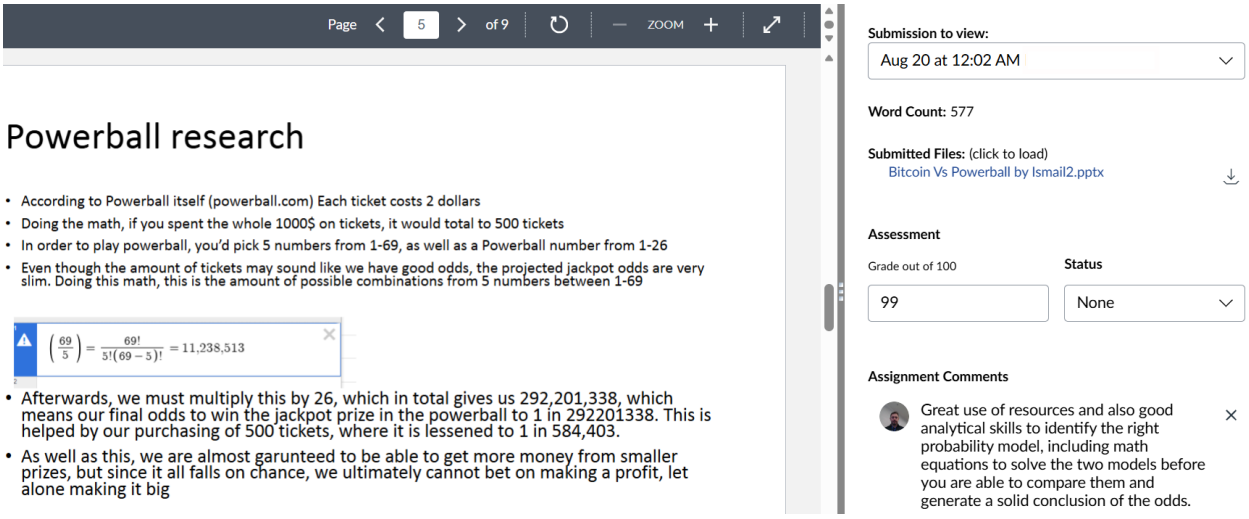
- Identify assumptions that could lead to inaccurate results.
- Determine possible mistakes programmers could make.
- Explain how those mistakes could be prevented.

FBLA Event: Computer Problem Solving

Students investigated and compared the probability of winning the Powerball lottery versus successfully mining Bitcoin. Students researched the problem, gathered supporting information, analyzed the mathematics required for comparison, explored probability concepts, and followed a design-thinking process to develop a proposed software solution before implementation.

Evidence Required

- One Graded Student Sample
- Teacher Feedback



AP Computer Science A

Technology Leadership Speaking Challenge

FBLA Event: Impromptu Speaking

Students receive a technology topic, are given twenty minutes to prepare, and deliver a five-to-seven-minute speech demonstrating communication, leadership, and technical understanding.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

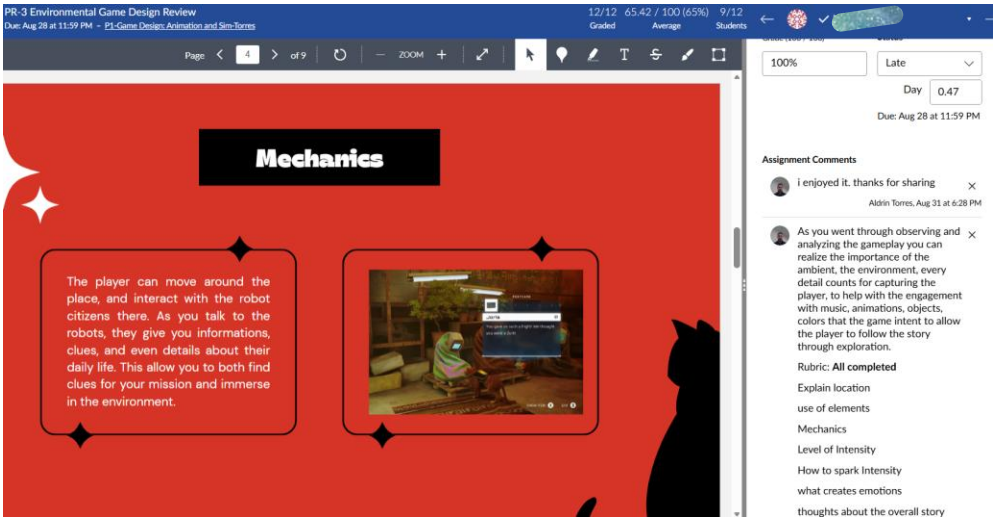
Game Design and Simulation

Environment and Emotional Design Analysis

Students evaluate how environmental design, audio, storytelling, color theory, level design, and gameplay mechanics affect player engagement and overall user experience.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback



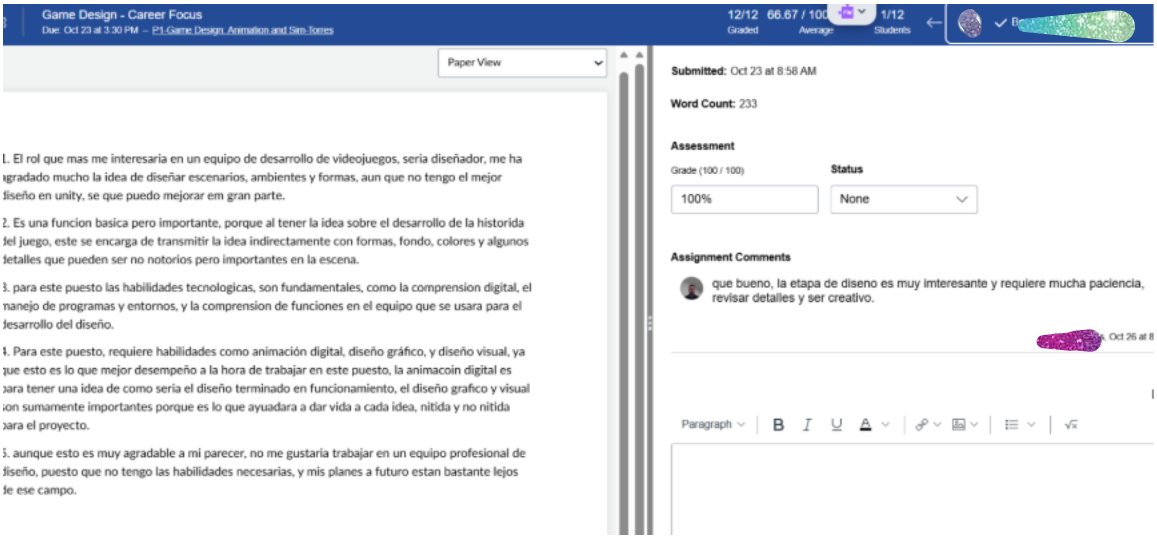
Industry Certification Evidence - Standard 1 FBLA Activities #2

Game Design & Simulation

Activity Summary: Careers in the Video Game Industry activity. Students explored game development careers, technology skills, and FBLA career connections while reflecting on professional roles within the game industry.

FBLA/Career Development Connection: Students explored career pathways, communication skills, workplace expectations, and employability skills aligned with FBLA competitive events and career readiness standards.

Student Sample Evidence and teacher Feedback

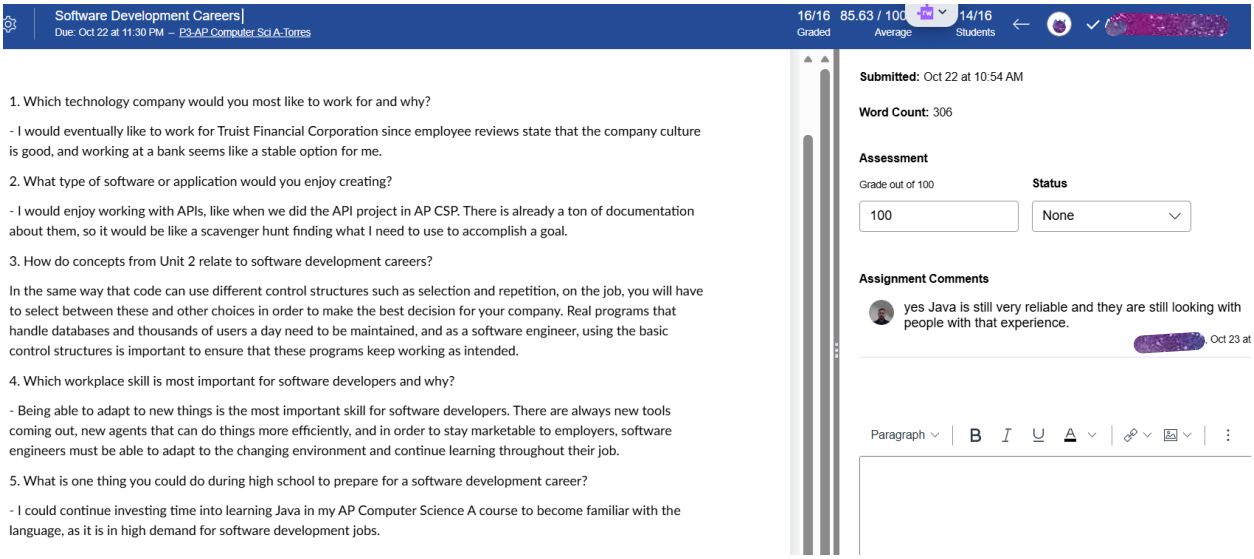


AP Computer Science A

Activity Summary: Students explored Java-related technology careers in Metro Atlanta and connected Unit 2 programming concepts to software development professions.

FBLA/Career Development Connection: Students explored career pathways, communication skills, workplace expectations, and employability skills aligned with FBLA competitive events and career readiness standards.

Student Sample Evidence and Teacher Feedback

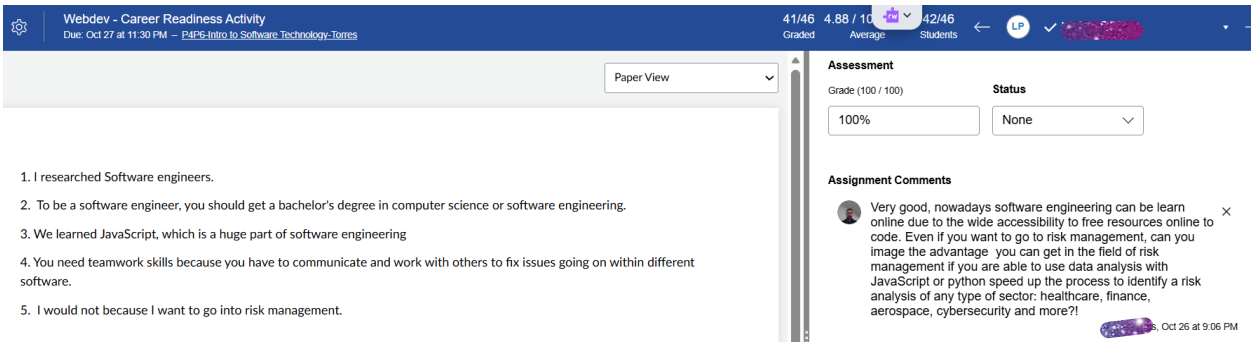


Introduction to Software Technology / CSP

Activity Summary: Students reflected on their Career Exploration Website project and connected technology careers, certifications, and FBLA employability skills.

FBLA/Career Development Connection: Students explored career pathways, communication skills, workplace expectations, and employability skills aligned with FBLA competitive events and career readiness standards.

Student Sample Evidence and Teacher Feedback:



AP Computer Science Principles

Activity Summary: Technology Careers in Metro Atlanta activity. Students researched data, AI, cybersecurity, and technology careers and connected course content to workforce opportunities.

FBLA/Career Development Connection: Students explored career pathways, communication skills, workplace expectations, and employability skills aligned with FBLA competitive events and career readiness standards.

